



Analyzing the Iranian Digital Pulse

Sentiment & Topic Trends on X (Twitter) 2022–2026

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Why Iran? Why Now?

Digital discourse as a critical "vent"

- **Closed Society:** X (Twitter) is the primary outlet where traditional polling fails.
- **Mood Tracking:** Real-time monitoring of social and political shifts.
- **Unfiltered Insight:** A direct look into a society under heavy media control.



Methodology — Building the Pipeline

Multi-model validation for high precision

Active Learning

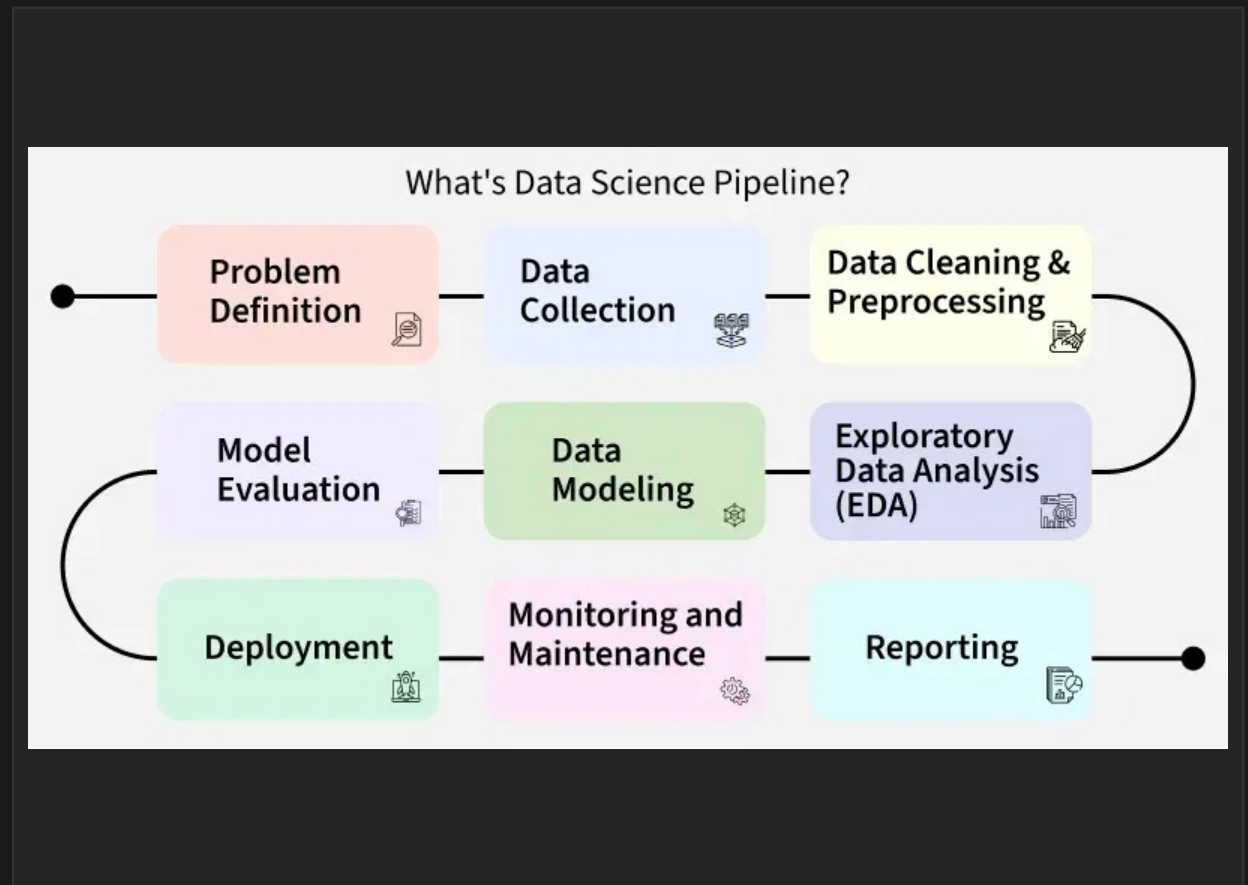
Iterative human-in-the-loop validation to refine model performance.

Logistic Regression

Used as a **robust baseline** for reliable sentiment and category prediction.

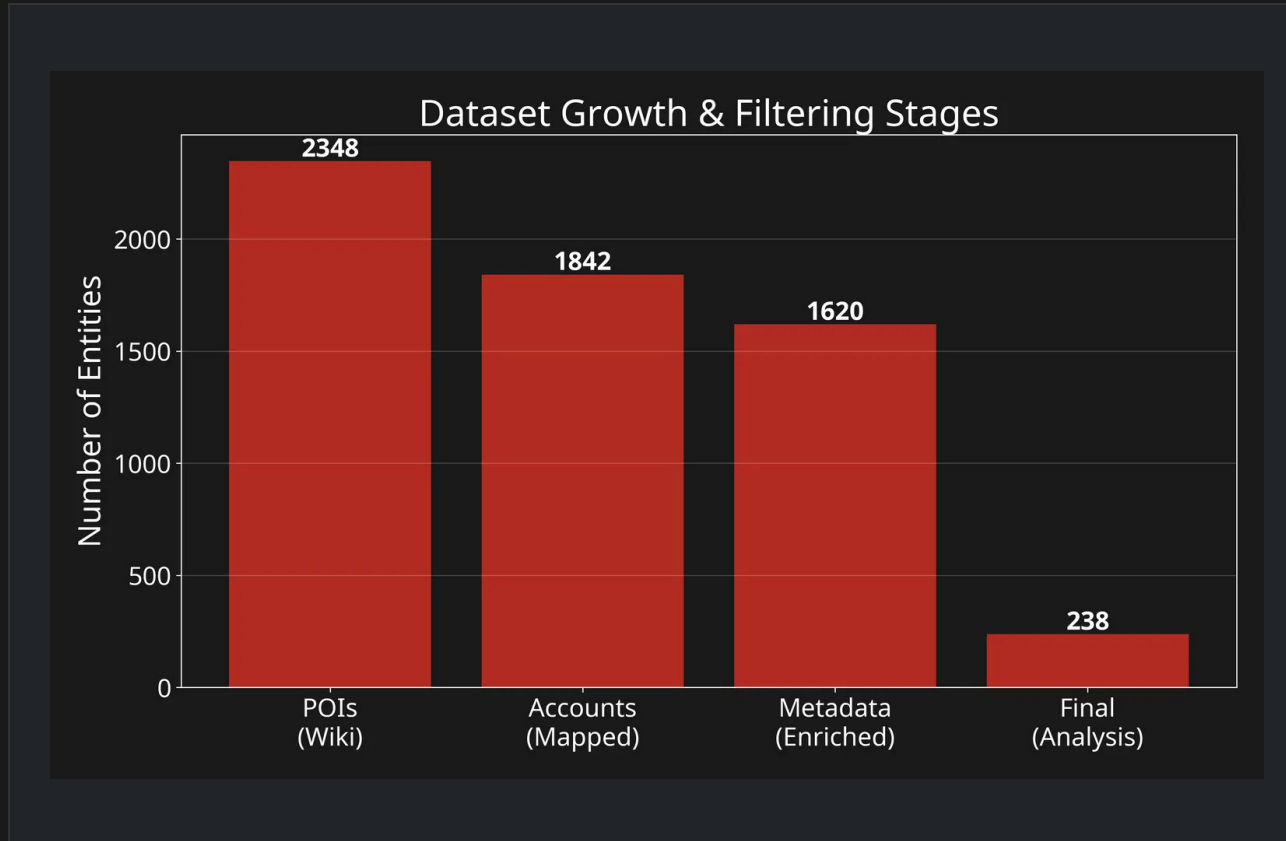
Persian-BERT

Advanced transformer models for deep contextual and semantic analysis.



Dataset Collection & Filtering Stages

From 2,348 POIs to 238 core verified users



Stage 1: POI Collection

2,348 entities across 8 categories collected from Wikipedia & Wikidata.

Stage 2: X Mapping

1,842 accounts successfully mapped to POIs with active metadata.

Stage 3: Analysis Set

238 core users selected for in-depth sentiment and topic analysis.

Feature Engineering

Training models on 300+ features across 3 stages

01

Textual Features (TF-IDF)

Unigrams and Bigrams extracted from **User Bio and Display Name**. Focus on identity markers.

03

Contextual (Persian-BERT)

768-dimensional **SBERT embeddings** capturing deep semantic nuances and Persian slang.

02

Numerical Metadata

Follower count, Following count, Statuses count, Bio length, and **Follower/Following ratio**.

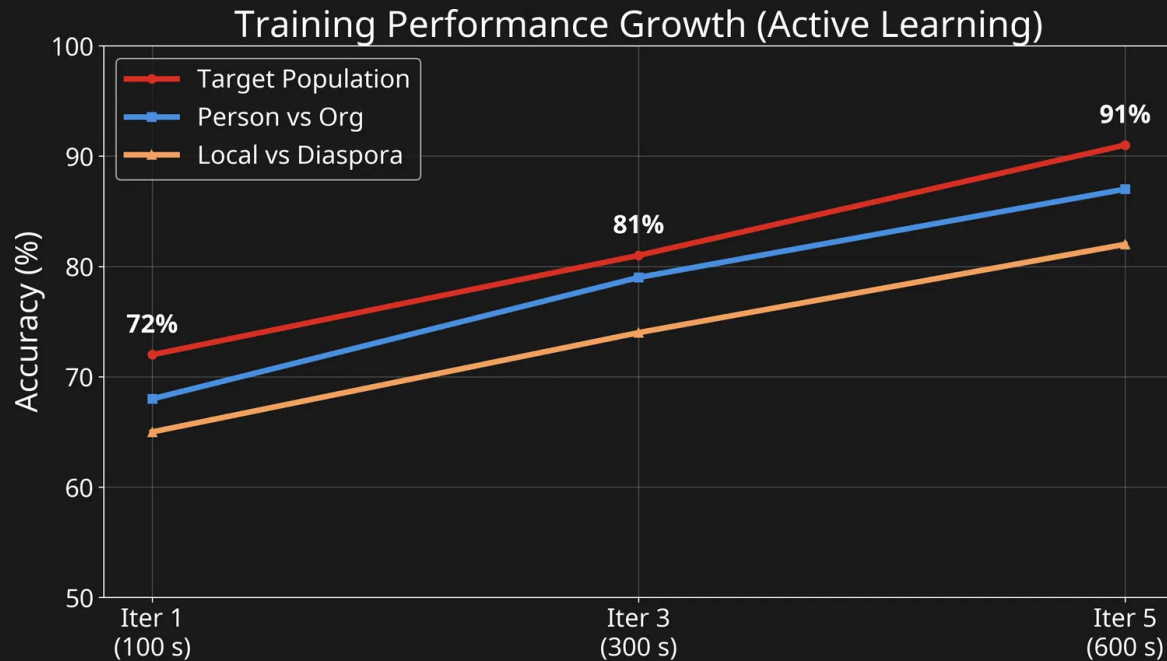
04

Topic (BERTopic)

Dimensionality reduction (UMAP) and **Clustering (HDBSCAN)** for automated topic discovery.

Training Performance Growth

Step-by-step improvement over 5 iterations



Iteration 1: Baseline

72% Accuracy on 100 random samples.
Established core feature sets.

Iteration 3: Mid-point

81% Accuracy after 300 uncertainty-sampled examples.

Iteration 5: Final

91% Accuracy for Target Population.
Production-ready performance.

21,810 Tweets — The Dataset

A robust and verified sample of
Iranian discourse

Our analysis is built on a high-quality dataset of
21,810 tweets collected between 2022 and early
2026, covering both local residents and the
influential diaspora.

21,810

TOTAL TWEETS
COLLECTED

93%

VALIDATION
ACCURACY

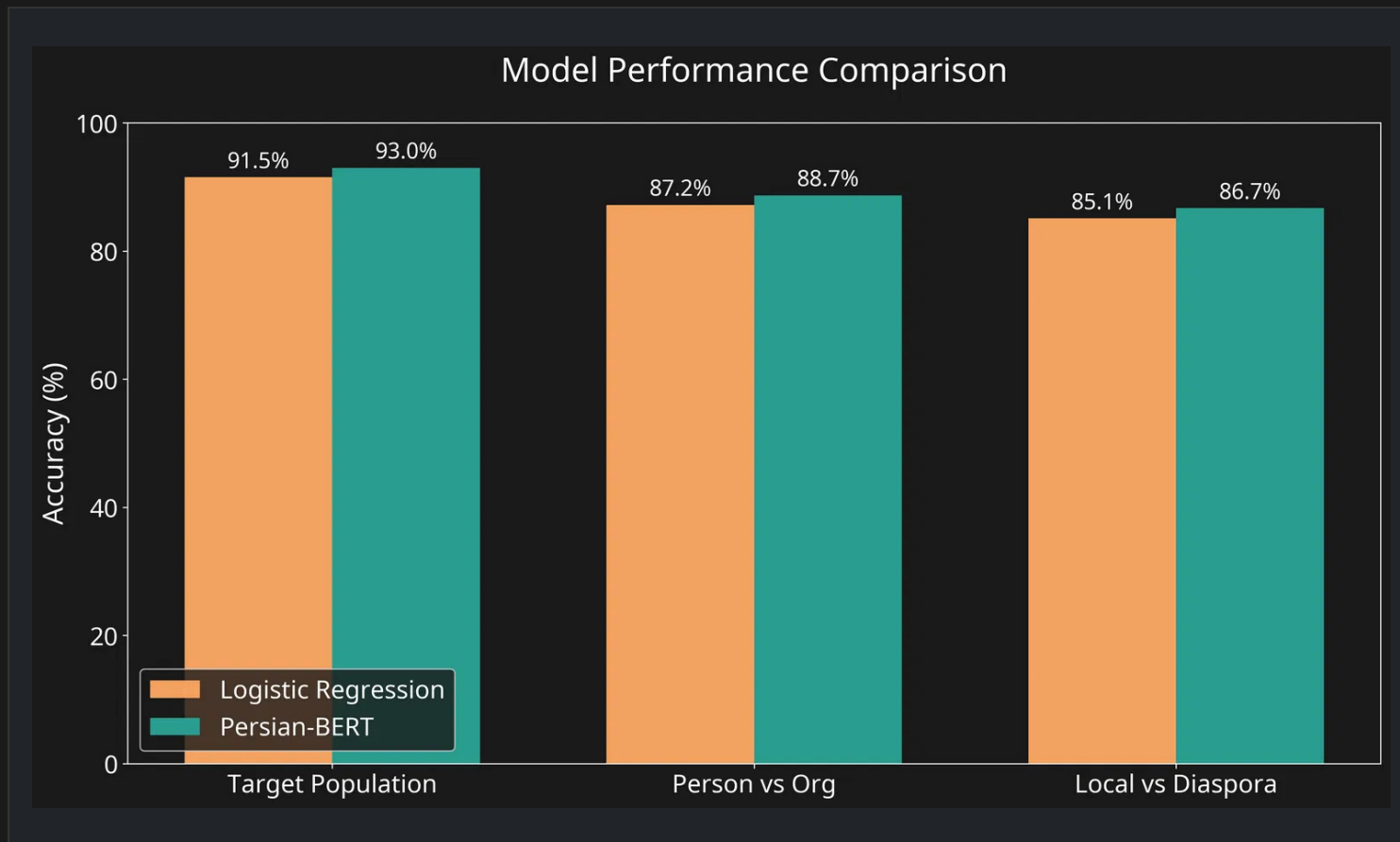
238

VERIFIED ACTIVE
USERS



Model Performance Comparison

Logistic Regression vs. Persian-BERT across 3 classification tasks

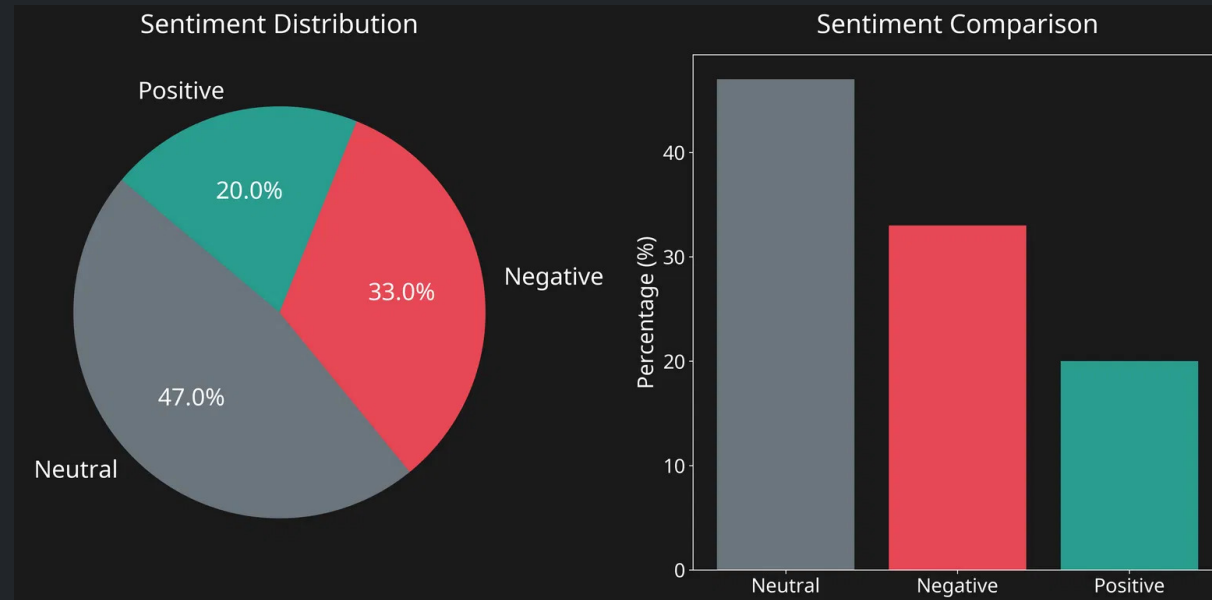


Logistic Regression: Serves as a highly interpretable and robust baseline, achieving over 91% accuracy in population identification.

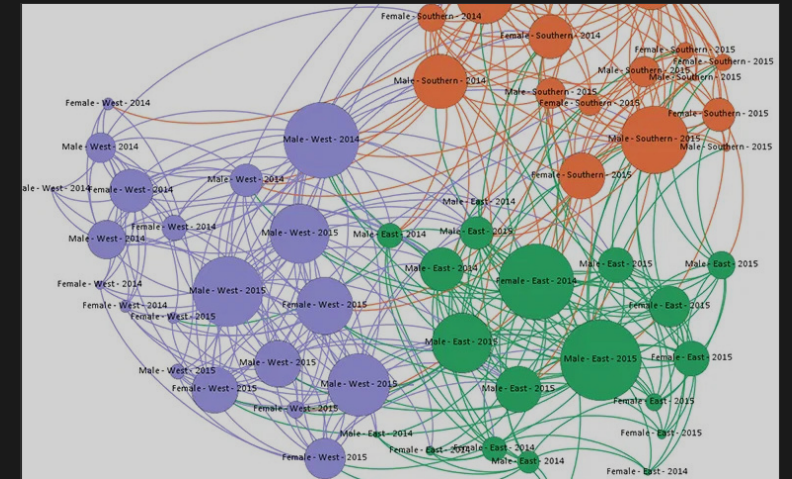
Persian-BERT: Provides state-of-the-art performance by capturing deep semantic nuances in Persian text, outperforming the baseline by ~2%.

Sentiment Analysis — A Dominant Tone of Tension

Negative sentiment consistently outweighs positive



47% Neutral | **33% Negative** | 20% Positive



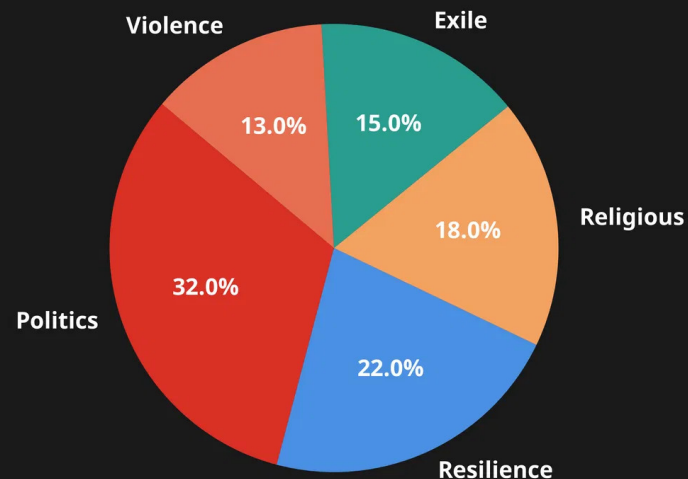
Network visualization of sentiment propagation across X.

Key Insight: Significant spikes in negativity were observed during **Jan-Feb 2026**, correlating with renewed protest activity and social unrest.

Conversation Topics

Top 5 identified topics in Iranian X discourse

Identified Conversation Topics (BERTopic)



Politics & Foreign Relations (32%)

Keywords: Nuclear, Sanctions, USA, Israel, JCPOA.

Personal Life & Resilience (22%)

Keywords: Family, Hope, Daily Life, Art, Poetry.

Religious Discourse (18%)

Keywords: Faith, Ramadan, Clergy, Tradition.

Exile & Opposition (15%)

Keywords: Diaspora, Protest, Freedom, Pahlavi.

Violence & Social Unrest (13%)

Keywords: Arrests, Executions, Justice, Mahsa Amini.

Event & Sentiment Timeline

Correlating political events with digital sentiment

Sept 2022

Mahsa Amini Protests

Sharp spike in **Negative Sentiment** and "Violence" topics. Discourse shifted to systemic critique.

Early 2023

Economic Sanctions Pressure

Increased discussion on **Sanctions and Inflation**. High prevalence of "Disgust" emotion.

Late 2024

Regional Tensions

Spike in **Fear and Anger**. Focus shifted to foreign policy and security concerns.

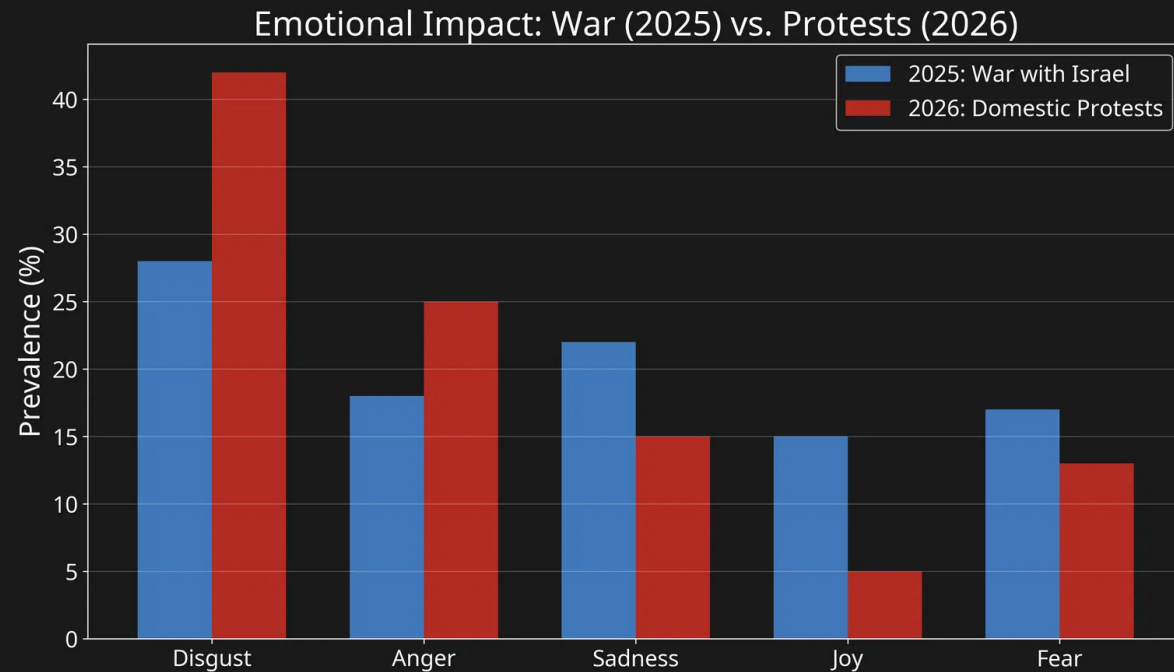
Jan-Feb 2026

Renewed Social Unrest

Highest levels of **Disgust and Violence** topics recorded. Significant digital mobilization.

2025-2026: War & Domestic Unrest

Emotional manifestations of geopolitical and social crises



2025: War with Israel

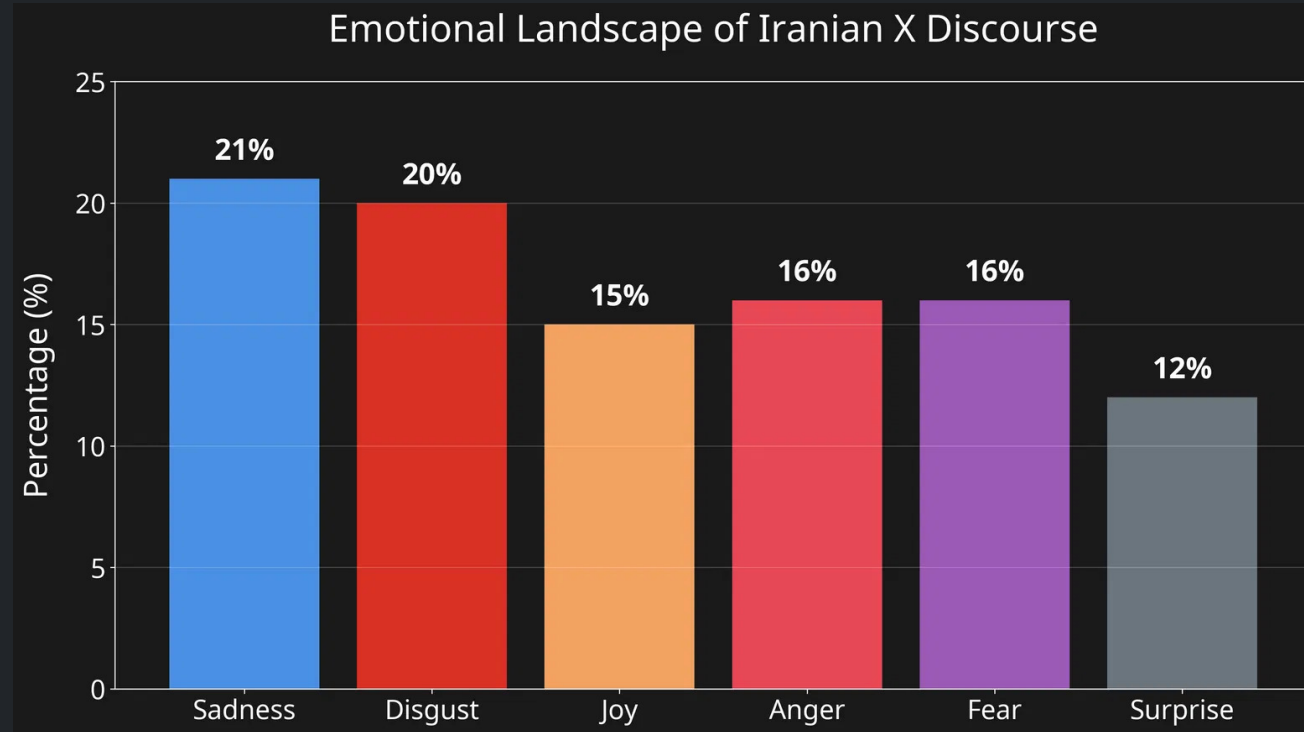
Characterized by sustained **Disgust** and **Sadness**. Digital discourse focused on national security, regional instability, and fear of escalation.

2026: Domestic Protests

Marked by extreme spikes in **Disgust** and **Anger**. "Joy" levels dropped to record lows as public discourse shifted toward systemic internal grievances.

Disgust & Sadness Rule

The Emotional Landscape of Iranian X Discourse

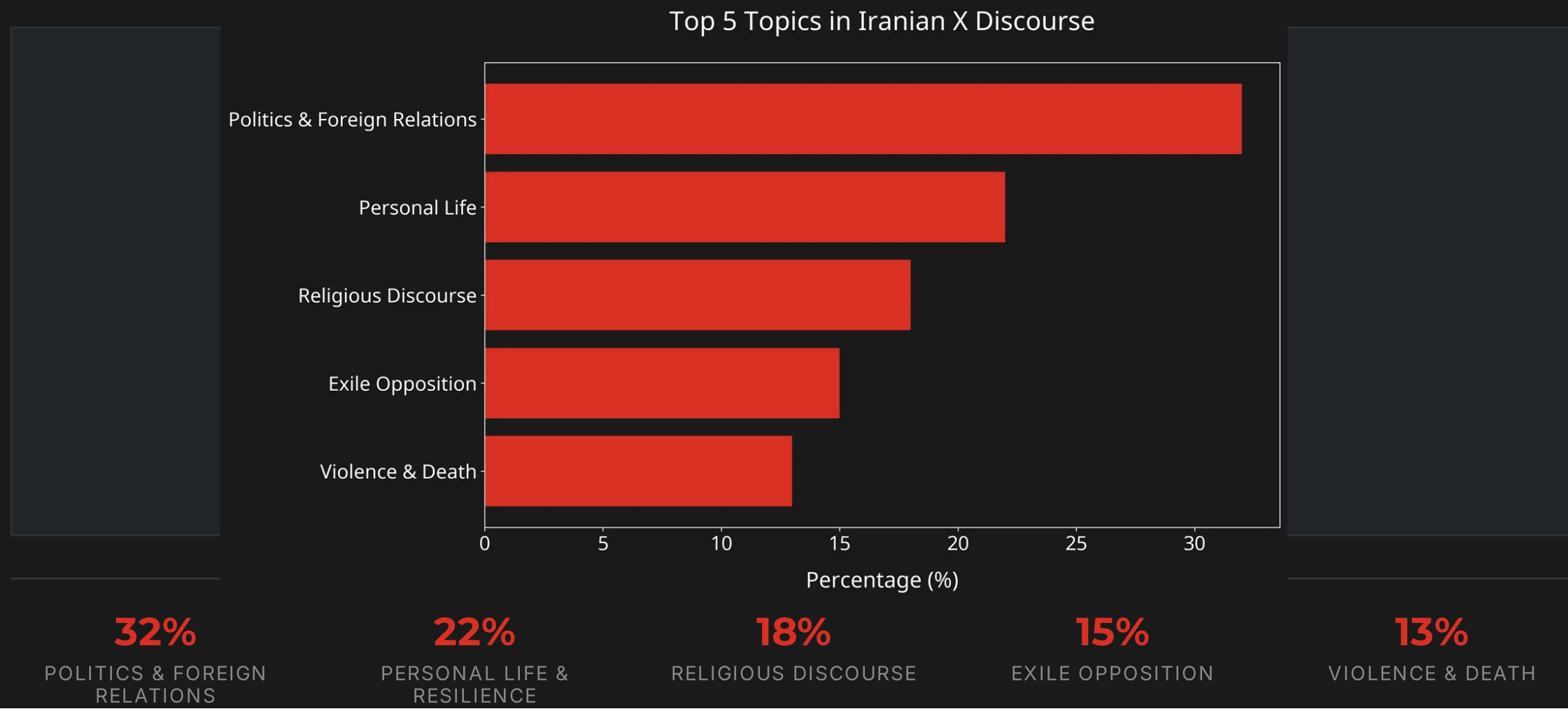


Sadness (21%) & Disgust (20%): These dominant emotions reflect a deep collective response to social restrictions and political events.

Emotional Gap: Joy (15%) is largely confined to personal life, showing how politics shapes the public emotional pulse.

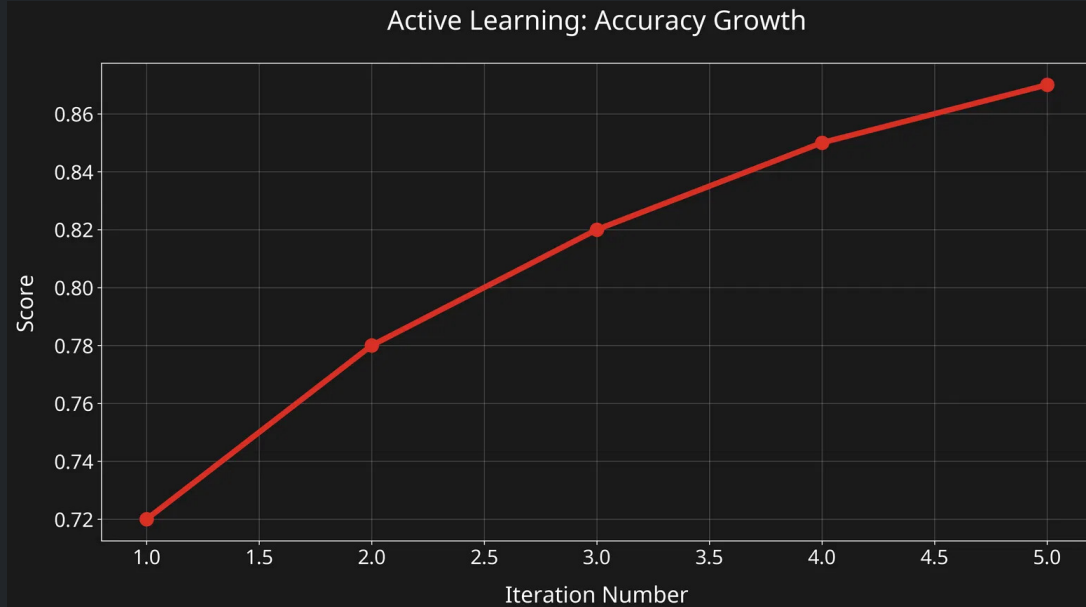
Politics & The West — What Are They Talking About?

Top 5 Topics in Iranian X Discourse (BERTopic Analysis)

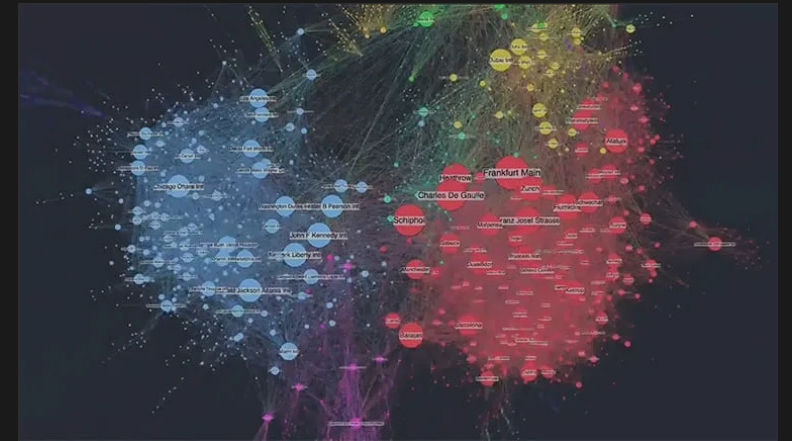


Active Learning Performance — Efficiency in Labeling

Accuracy jumped from 72% to 87% with minimal manual effort



Iteration 1: 72% | **Iteration 5: 87%**



Validation of network nodes through human-in-the-loop labeling.

Key Insight: Uncertainty sampling focused human effort on the "hardest" cases, achieving production-level results with only **500 manually labeled examples**.

Politics Shapes Emotion

The Core Dynamic of Iranian Digital Discourse

01

Direct Correlation

Periods of high political tension and violence show a **near-perfect correlation** with spikes in Disgust and Sadness.

02

Emotional Resilience

Cultural, religious, and personal topics act as a **buffer**, maintaining a baseline of Joy even during political crises.

03

The Digital Weather Vane

X (Twitter) discourse serves as an accurate **real-time predictor** of the broader social mood in Iran.

"Digital discourse is not just noise; it is the pulse of a society in transition."

Conclusions — What We Learned

Key insights from the Iranian digital narrative

01

Non-Monolithic Discourse

The Iranian digital space is not just about politics; **personal resilience** and cultural life persist even during times of high tension.

02

Emotional Dynamics

Negative emotions like **Disgust and Sadness** are the primary collective responses to political violence and social restrictions.

03

Methodological Success

The combination of **Logistic Regression** and Persian-BERT provides a robust, scalable framework for analyzing complex Persian text.

04

The Diaspora Pillar

The Iranian diaspora remains a **crucial pillar** of the narrative, often amplifying local voices and shaping international perception.

Limitations & The Future

Expanding the window into closed societies

Methodological Value

Data science provides a **unique window** into societies where traditional polling and media are restricted.

Platform Expansion

Future research will integrate data from **Telegram** and **Instagram**, which are heavily used within Iran.

Advanced NLP

Leveraging **Persian BERT** for even deeper contextual and semantic analysis of slang and regional dialects.

Deep Learning Pipeline

